ACHIEVING GOOD CORPORATE GOVERNANCE USING A FINANCIAL OPTIMISATION MODEL:

MANAGERIAL AND FINANCIAL ACCOUNTING PERSPECTIVES

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1. Aims and Context of the Project

1.1. Background

The outraged corporate collapses which are following the phenomena of corporate failure over decades around the world (Clarke, F, Dean & Oliver 2003) have urgently required good corporate governance (GCG) practice around the world. Nevertheless, the fact is history has been repeated and responses have been made but the failures continue. Indeed, this shows how fragile and volatile the global economy is. Many companies have realised the essential need for good corporate governance and sound financial management strategies, nonetheless, why is there still company failure and why is it that a company which claims to have implemented corporate governance still cannot achieve value maximisation? Indeed, the main question is: what does *good* mean in both good corporate governance and good financial management? What *good strategies* need to be implemented by a company in managing its business? How to produce these good strategies and how a quantitative model, i.e. accounting optimisation model, plays its role in producing such strategies?

Despite the fact that implementing a real sound corporate governance system is the answer, currently, corporate governance practice is still viewed as a risk for a company and costly (Dallas 2004; Fabozzi & Modigliani 2009) and therefore many companies just simply apply tick-boxes corporate governance practice (HIH Royal Commission 2003). Literature shows that the current management strategy, using managerial, financial and accounting perspectives, implemented by companies to achieve GCG is still partial, due to, for example, not integrating all the perspectives of or including the business and regulatory environments (Ho & Lee 2004). As a result, the partial strategies will not be appropriate for becoming the basis of a management decision making process to achieve GCG. Therefore, implementing all theoretical GCG concepts into corporate system and management activities in integrated ways, and synchronising the concepts with the corporate objectives and management activities would benefit the company thereby increasing the value of the company and making the company sustainable (OECD 2004, 2009). In addition, the abstract and normative corporate governance concepts need to be quantified and derived into an integrated corporate financial model. The integrated financial model needs to reflect all business environments and risks such as the volatility of economy which company faces so that management can do risk management activities thereby minimising the costs while maintaining the company's value creation activities. The integrated quantitative model which reflects corporate governance practice of the company is urgently needed to monitor the success of GCG practice which can be measured by a company's value.

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1.2. Aims of the Project

Sound financial strategies are essential to achieve GCG. A financial model inclusive of corporate governance practice and business perspectives, and considering risks and regulatory environments such as accounting practices, financial market and capital market characteristics is important. The main objective of this research is to develop a sound financial model using optimisation approach to measure and manage the value of a company for achieving GCG. The output of this study will be sound financial management strategies which incorporate corporate governance practice, business perspectives and the dynamic of the economy. The specific aims of this research are to:

- determine GCG practice, business risks and regulatory environments that should be incorporated into the financial management strategies of a company;
- quantify the policy and mechanisms above, and identify the right proxy for them;
- design an integrative dynamic multi-period financial optimisation model for sound financial management strategies;
- simulate the model to obtain sound financial management strategies and achieve GCG;
- measure and manage value of a company using the integrated financial optimisation model.

1.3. Contribution to Knowledge

Literature on corporate governance and corporate finance has emerged extensively but has overlooked a quantitative model which can measure the normative corporate governance concepts and assess the effectiveness of GCG practice by measuring and managing the value of a company. While there are corporate financial models, the models are still partial since they do not incorporate corporate governance and regulatory environments such as accounting practices, financial and capital market characteristics. In this regard, it is potential for a mathematical model, i.e. an optimisation model, to fill the need for an integrated quantitative model in the area of corporate governance. To the author's knowledge while an optimisation model has been widely used for decision making process, an integrated financial optimisation model for achieving GCG has not been done before. Literature shows that the application of an optimisation model mainly in the area of capital budgeting, project management, investment, and production and inventory planning, etc. (see Gass & Harris 2001; Anderson, Sweeney & Williams 2005; Ragsdale 2007).

Therefore, the major contribution of this paper to knowledge is offering an integrated financial optimisation model to achieve GCG practice. The integrated financial model will

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quantify and derive the normative and qualitative characteristics of corporate governance practice and regulatory environments which a company faces. Hence, the output will be integrated financial management strategies to guide the managers to measure and manage the value of a company for achieving GCG practice. Using an optimisation approach, thereby, this study fills a gap in financial management literature as well as corporate governance so that companies can have sound financial model to produce good financial management strategies to achieve the advantages of GCG.

1.4. Introduction to the Context of the Project

The global and volatile economy together with other business environments have been viewed as risks for a company since they influence the success of a company in creating its value (Dallas 2004). Corporate governance takes these risks into consideration in its broader concept: "the entire network of formal and informal relations involving the corporate sector and their consequences for society in general" (Keasey, Thompson & Wright 1997, p. 2). Corporate governance suggests all risks including regulations and other external systems controlling a company such as self-regulation system and 'best practice' norms should be embedded in the centre of corporate structure (Farrar 2008; Iskander & Chamlou 2000). The necessary regulatory environments, known as external governance mechanisms, are specifically related to financial management strategies, such as the market for corporate control, accounting regulation and securities law, diversification requirements related to ownership of institutional investors and financial intermediaries, capital and financial markets characteristics, and industry practice (Bushman & Smith 2001).

Apart from these external risks, a company also faces internal risks which actually in some extent reflect the external risks. They are embedded in the policies of its main activities. These internal policies are called, by Bushman and Smith (2001, p. 238), corporate control mechanisms, "the means by which managers are disciplined to act in the investors' interests". Allen and Gale (2000) identify these control mechanisms as corporate governance instruments or provisions which consist of internal corporate governance mechanisms. The relevant internal policies are those related to finance, investment and accounting, which consist of, for example, managerial incentive plans, capital structure, risk and return, asset management and dividend growth policy. Specifically, financing policy made by the management in order to find financing for its activity whether by issuing stocks or debt instruments affects the company's value. Similarly, investment policy, which is related to internal investment such creation of assets and plant/market expansion and external investment by buying the other company's stocks, affects the main goal of a company to

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maximise the owners' wealth. The accounting practice and standards adopted by a company affect the presentation of monetary information of the financing and investment activities in the financial statements.

The description above suggests, to achieve good corporate governance practice and to maximise company value, management should accommodate, incorporate and synchronise all those external and internal factors into its corporate strategy and quantify them into financial strategy. Accordingly, an integrated financial modelling approach is essential for management to create sound financial strategies: strategies that are able to reduce risks and enhance firm value hence achieving the objective of GCG.

This study proposes sound financial management strategies using an integrated financial model to measure and manage the value of a company for achieving GCG. The financial model will be developed using an optimisation approach and accommodate CG internal and external instruments in integrative ways. The proposed model is more than a financial model as it integrates corporate governance practice and regulatory environments specifically accounting practices, and capital and financial market characteristics, and can be used for valuation of the company.

1.5. Statement of Significance

Current corporate collapses and financial crises suggest that, while the weakness of corporate governance is the cause, good financial strategies are crucial to achieve GCG. In fact, the current financial crisis has called for sound financial strategies to be implemented by companies around the world. An integrated financial model which considers business risks i.e. external regulatory environments as well as internal corporate policies is essential since it provides management with better equipment to produce sound financial strategies for achieving the advantages of GCG.

The integrated financial model operationalises the qualitative advantages and constraints of GCG practice, and regulatory environments which are derived from the theory and literature discussion into quantitative dimensions. Therefore, management can create good financial strategies by measuring the optimal value of corporate governance implementation given all internal policy and external regulatory environments as the constraints. Indeed, this would be of significant value to organisations pursuing best corporate governance practice as well as complying with the regulations. In addition, from an academic point of view, this type of research would fill the vacuum of an integrated financial optimisation model which is potentially applied in corporate governance study. An integrated

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optimisation model offers a significant contribution to the literature because it provides a managerial application model showing interactivity amongst individual governance variables as well as considering external governance mechanisms. The model would be potential for company valuation so that by integrating all discipline areas, it extends further the capital market and corporate governance research.

2. Literature Review and Conceptual Framework

2.1. An Interrelationship between Corporate Governance, Capital Market and Accounting

Many theories shaped corporate governance such as agency theory, managerial hegemony theory, stewardship theory resource dependency theory, institutional theory and stakeholder theory (Clarke, T 2004; Lawrence & Lorsch 1967). Based on these theories, internal and external governance instruments are developed to ensure that all of the external and internal parties participate and work together in order to achieve value that fits all stakeholders' interests: the sustainable value, and theoretically, the benefits of GCG as follows can be achieved: "1) reduce risk; 2) stimulate performance; 3) improve access to capital markets; 4) enhance marketability of product/services by creating confidence among stakeholders; 5) improve leadership; 6) demonstrate transparency and accountability" (Collier & Agyei-Ampomah 2007, p. 84). The concept of GCG when it is applied properly, will mitigate the conflict of interests thereby adding value to a company and gives a positive signal to the market as shown by the study of Gompers, Ishii and Metrick (2003, p. 107), and ensure the achievement of economic benefits by making the use of resources of a company efficient thereby underpinning growth and increasing company performance (Brown & Caylor 2009; OECD 2004).

The internal instruments of corporate governance are developed to fulfil responsibility to the stakeholders; and under stakeholder theory, they reflect a company's activities in considering and accommodating the external demands. Similarly, based on agency theory, internal governance mechanisms are developed to alleviate the agency problem. The theory argues that the separation of management and the financing of business entities gives opportunities for management not behaves on the shareholders' behalf thereby creating agency costs (Berle & Means 1933). It creates information asymmetry that management has more information than the shareholders and other parties. To alleviate the agency problems and to fulfil responsibility to the stakeholders, GCG ensures that the checks and balances procedures take place. This can be achieved by implementing the internal governance

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mechanisms such as: 1) empowerment of the organisation structure (e.g. requiring high responsibilities of the board, strengthening the supervisory functions and ensuring that other functions in every stage of decision making are effective and efficient); 2) make the financial reports which reflects the business operations become reliable; and 3) ensure the company complies to the regulations (Calder 2008; Dietl 1998; Jensen & Meckling 1976; OECD 2004; Rezaee 2007, p. 50). This implementation would make the business operations become more effective and efficient thereby minimising the agency costs: signalling, screening, monitoring and residual costs. These internal governance mechanisms in organisation structure would be applied in term of, for example, size of the boards, composition of the boards, leadership of the boards, size of the company, ownership structure, board committees: audit committee, remuneration committee and corporate governance committee (Adams & Mehran 2003; Barnhart & Rosenstein 1998; Berghe & Levrau 2004; Gillan, Hartzell & Starks 2003; Hermalin & Weisbach 1988; Rezaee 2007).

In addition to the organisation structure, some financial and accounting policies which are part of financial and risk management strategies are also developed as internal governance instruments. These internal company's policies accommodate external governance mechanisms such as industrial codes of conduct, professional practices, regulations and other business environments. The example of this is leverage or capital structure which is an important tool to protect the shareholders and manage the conflict of interest between shareholders and debt-holders. Similarly, a dividend policy is also a governance tool that in some countries such as in the US, companies adopt non-dividend policy since the regulation requires that whereas in some other countries dividend is viewed as a signal for a healthy company. Accounting policies of the company are also the company's internal mechanism which captures the industry regulatory and accounting practices (standards).

As the company has good internal governance mechanisms, it can tackle business risks and becomes sustain. Accordingly, capital market as one of the external corporate governance mechanisms will then appreciate these good internal governance mechanisms (Rezaee 2007). The capital market captures the accounting and finance information sent by the company and responds to this information by reflecting in adjustments of the company's market share price. In this relationship, accounting has an important role since it measures all business activities including corporate governance practice in monetary units and sends to the market through accounting and financial data. As shown by literature, market pays higher premium for companies which implement GCG practice (Beiner et al. 2004; Black, Jang & Kim 2006 ; Brown & Caylor 2009; Gompers, Ishii & Metrick 2003; OECD 2004). The historical price

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movement of failure companies explained how market had responded negatively for these failure companies (Clarke, F, Dean & Oliver 2003).

2.2. Literature Gap

The interrelationship between GCG, capital market and accounting is clear. However, the problem is how to apply the internal mechanisms and accommodate the external governance mechanisms effectively and optimise their functions which at the end adds value of a firm. This question also leads to the discussion that while GCG promises many economic benefits, there are still corporate collapses and incongruously some of those failure companies claimed have GCG (Clarke, F, Dean & Oliver 2003). The "tick-boxes" practice of a collapsed company such as HIH (HIH Royal Commission 2003) gives a learning point that qualitative recommendation of GCG principles is not enough.

One is missing from the discussion of GCG is a quantitative financial model of corporate governance which quantifies the qualitative recommendations of GCG best practice. A sound financial model which captures and measures the monetary effects of all internal and external corporate governance practice is important so that the corporate governance practice can be evaluated and measured its economic benefit in term of monetary units. The potentiality of a mathematical model, i.e. an optimisation model, would fill the need for a quantitative model for corporate governance. To the author's knowledge, this has not been done before. Literature shows that the application of an optimisation model mainly discusses partial topics of financial management and does not integrate business environments and corporate governance practice of the company (see Gass & Harris 2001; Anderson, Sweeney & Williams 2005; Ragsdale 2007). The existing financial models in these four decades such as Ijiri, Levy and Lyon (1963), Carleton (Carleton 1969, 1970), Warren & Shelton (1971), Hamilton and Moses (1973), Carleton, Dick and Downes (1973), Merville and Tavis (1974), Ho and Lee (2004), and Shahnazarian (2004) have overlooked some important points which cause them not reliable for decision making and, particularly, achieving GCG. The limitations of these existing models are:

1) The type of the current financial models are more static model which do not reflect the real environments of which a company faces such as dynamic of the economic condition and the uncertainty of the future.
2) The existing models neither consider current practice of GCG nor risks which influence a value of a company.

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3) The existing financial models contain 'accounting noises' since use accounting information as input for the models. This might cause the reliability of the outcome for sound decision making analysis.
4) While the current financial models are developed based on accounting information and financial statements, they do not show clear and correct the complex interrelationship between accounts of financial statements. In addition, the model also lack of justifications of the equations of the objective function and the constraints.
5) The existing financial models do not consider about any earnings quality such as predict ability. They only give some guidance for management how to optimise the firm value.
6) The current financial models do not have sufficient robustness tests which might decrease the quality of the model.

2.3. Conceptual Framework

This study is based on the postulate of capital market research that market will respond on all publicly information depending on the efficiency of the market. Market responds on accounting information and public information which conveys all company's activities including corporate governance practice. The conceptual framework of this study further is depicted in Figure 1. It suggests that corporate governance requires good financial management strategies and a quantitative model as a tool of management to produce the strategies, and manage the company's value. To create sound financial strategies, it is important to consider both company's corporate governance practice and regulatory environments related to accounting, and financial and capital markets.

Specifically, the company's corporate governance practice is related to financial policy. Since financial activities are related to investment policy, the model also considers this policy. Next, as accounting plays an important role as a medium between management and outsiders (Bushman & Smith 2001), it is important also to take accounting policy into the model. As argued by Ho and Lee (2004), empirical evidence suggests that financial managers make decisions rooted in financial statements due to the informational content in financial reporting.

In addition to the corporate governance practice variables, some related factors of regulatory environment will be included in the model. Specifically, the regulations will be related to accounting, and capital and financial market characteristics, and other diversification requirements related to these market characteristics (Dietl 1998). Finally, by integrating these variables, the integrated financial optimisation model will have an output of

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sound financial management strategies based on specific corporate governance practices and regulatory environments which in turn achieve GCG and manage the value of the company.

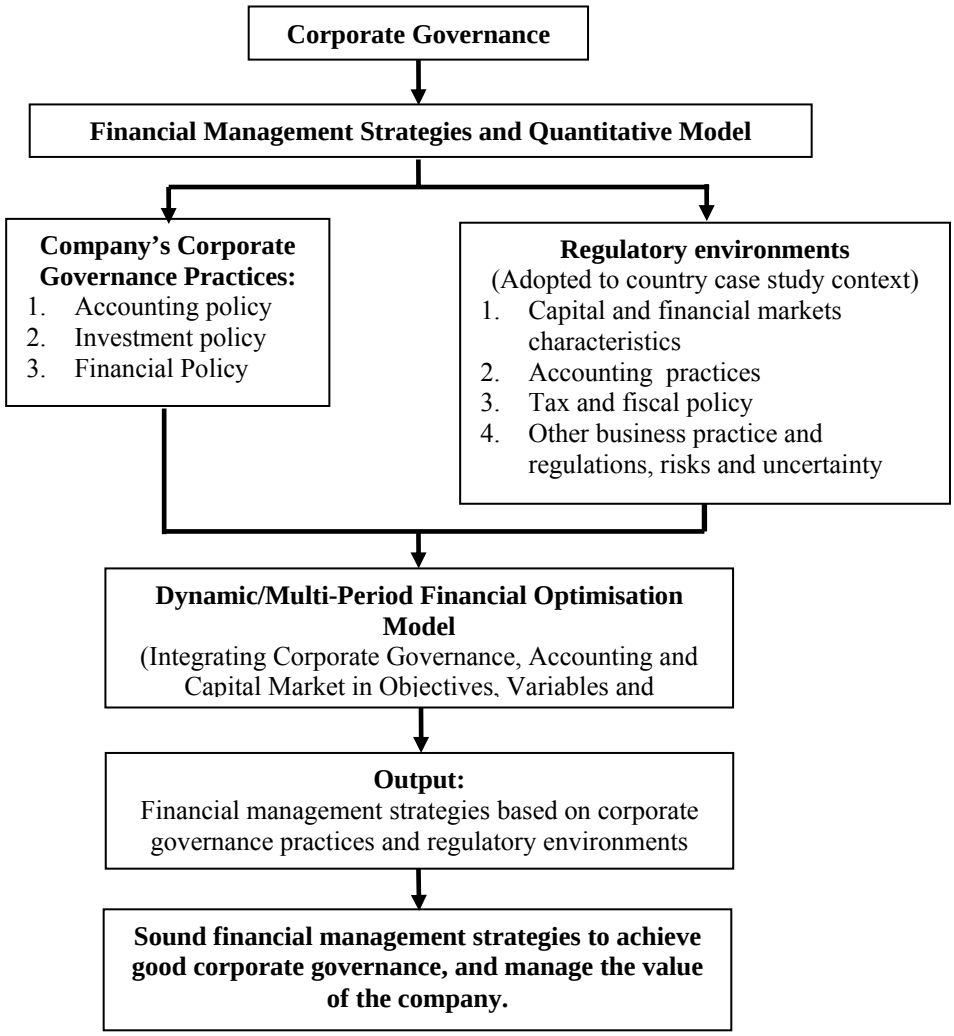


Figure 1. Conceptual Framework of An Integrated Financial Model using Optimisation Approach for Achieving Good Corporate Governance

3. Introduction to an Optimisation Model

To achieve GCG and obtain sound financial management strategies, this study develops a financial optimisation model. Giving extensions to the previous studies such as Carleton, Dick and Downes (1973), Ho and Lee (2004) and Lee, Lee and Lee (2009), an optimisation model which will be developed will articulate GCG practice and have regulatory environment variables as discussed above, namely: accounting practices and regulations, diversification requirements, capital and financial market characteristics, and industry practices. The model defines the objective function as maximising GCG practice, proxied by enterprise value measures. The objective function will have long horizons since, generally, financial planning is determined over multiple periods (Ho & Lee 2004).

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The constraints of the model will be developed from market frictions and selected regulatory environments as previously discussed. The examples are restrictions on the level of debt (leverage) and excessive cash positions, operating policy and liquidity policy. In addition to this, the control variables of the model will be defined as a specification in a state-dependent multi-period context (Ho & Lee 2004). The example is dividend payment which depends on the prevailing net income.

3.1. A Financial Optimisation Model for Achieving Good Corporate Governance

The model is composed of an objective function and constraints. In this model, the goal of a company will be to achieve GCG which leads to maximising shareholders' wealth (Keown et al. 2006). Enterprise value based on discounted cash flow (DCF) will be the proxy (adjusted present value (APV) is applied in this paper; see Copeland, Koller and Murrin (2000), and Koller, Goedhart and Wessels (2005, 2010) for the discussion and the benefit of APV model). The model has several constraints, representing the companies' internal governance policies related to accounting, risk management, financing and investment activities. The objective function and constraints also reflect the corporate governance practice and regulatory environments. The summary of the model proposed by this study is as follows. The parameters are shown in the appendix.

Objective function:

$$\text{Maximise APV of FCF'}_0 = \sum_{t=1}^{T} \frac{\text{FCFF}_t}{(1+k_u)^t} + \sum_{t=1}^{T} \frac{\text{VTS}_t}{(1+k_u)^t}$$

where:

APVof FCF'_0 = adjusted present value of total FCF in period zero;

FCFF_t = free cash flows to the firm in period *t*;

VTS_t = value of free cash flows from debt financing or value of tax shield in period *t*;

k_u = unleveraged cost of capital.

Constraints:

A. Accounting Equation Constraints

$$\text{FCF}_t = \text{FCFF}_t = \text{CFO}_t + \text{Int}_t + \text{CFI}_t \quad (1)$$

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$$FCFF_t = \overline{Sales_t} - (AR_t - AR_{(t-1)}) + ORev_t - (Rec_t - Rec_{(t-1)}) - COGS_t - (Inv_t - Inv_{(t-1)}) + (AP_t - AP_{(t-1)}) - OExp_t - (Prep_t - Prep_{(t-1)}) + (Accr_t - Accr_{(t-1)}) - Tax_t + (TP_t - TP_{(t-1)}) + i_t(1-\tau) + CFI_t \quad (2)$$

$$FCFE_t = FCFF_t - Int_t + \Delta LTD_t - pdiv_t + \Delta cashPS_t \quad (3)$$

$$Int_t = i_t(1-\tau) \quad (4)$$

$$VTB_t = (i_t \, x \, \tau) \quad (5)$$

$$FCFE_t - (div_t + \Delta cashTS_t) + \Delta cashCS_t = Cash_t - Cash_{(t-1)} \quad (6)$$

$$Tax_t = \tau\,(EBT_t + \Delta T_t + \Delta P_t) + DTE_t \quad (7)$$

$$DsC_t = \frac{EBIT_t}{i_t + \frac{COD_t}{(1-\tau)}} \quad (8)$$

B. Accounting Policy Constraints

$$\Delta P_t \leq x\, exp_t \quad (9)$$

$$\frac{Sales_t}{\overline{AR_t}} \geq Avg \sum_{t=1}^{5} \frac{Sales_t}{\overline{AR_t}} \quad (10)$$

$$ARTO_t \geq \overline{ARTO_t} \quad (11)$$

$$ITO_t \geq \overline{ITO_t} \quad (12)$$

$$\frac{COGS_t}{Inv_t} \geq Avg \sum_{t=1}^{5} \frac{COGS_t}{Inv_t} \quad (13)$$

C. Corporate Governance Policy: Risk Management, Financial and Investments Policy Constraints

$$\frac{CA_t}{CL_t} \geq Avg \sum_{t=1}^{5} \frac{CA}{CL} \quad (14)$$

$$\frac{CA_t}{CL_t} \geq \frac{\overline{CA_t}}{\overline{CL_t}} \quad (15)$$

$$\frac{LTD_t}{(LTD + Equity)_t} \leq \frac{\overline{LTD_t}}{\overline{(LTD + Equity)_t}} \quad (16)$$

$$\frac{TD_t}{TA_t} \leq \frac{\overline{TD_t}}{\overline{TA_t}} \quad (17)$$

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$$DsC_t \geq \overline{DsC_t} \tag{18}$$

$$(div_t + \Delta cashTS_t) \leq CFO_t + CFI_t + \Delta LTD_t - pdiv_t + \Delta cashPS_t + \Delta cashCS_t - (Cash_t - Cash_{(t-1)}) \tag{19}$$

$$div_t - \alpha div_{(t-1)} \geq 0; \tag{20}$$

$$\alpha_t \geq \frac{Q_t}{Q_{(t-1)}} \tag{21}$$

$$CIE_t \leq MV_t - \frac{1}{\alpha_t} MV_t; \tag{22}$$

$$P_t \geq \sum_{t=1}^{\infty} \frac{E(EPS_t + \frac{CIE_t}{Q_t} - \frac{CFI_t}{Q_t})}{(1+k_d)^t} \tag{23}$$

$$PPE_t - PPE_{(t-1)} > 0 \tag{24}$$

$$\frac{\varphi Sales_{(t-1)}}{\varphi PPE_{(t-1)}} \geq \frac{\varphi Sales_t}{\varphi PPE_t} \geq \sum_{t=0}^{5} \frac{\overline{\varphi Sales}}{\overline{\varphi PPE}} \tag{25}$$

$$\varphi Sales_t \geq \overline{\varphi Sales_t} \tag{26}$$

$$CFI_t \leq CFO_t + \Delta LTD_t - pdiv_t + \Delta cashPS_t + \Delta cashCS_t - (Cash_t - Cash_{(t-1)}) \tag{27}$$

$$Inc_t \leq \rho\,(FCFF_t - FCFF_{(t-1)}) \tag{28}$$

$$0 \leq \rho \leq \frac{(FCFF_t - FCFF_{(t-1)})}{FCFF_t} \tag{29}$$

4. Proposed Methodology and Research Design

To develop sound financial management strategies, a quantitative financial optimisation method will be incorporated in this project. Using a quantitative approach, a mathematical model will be built by making use of quantitative data of accounting, finance and capital market. This project will utilise a multi year dynamic optimisation model (Cornuejols & Tütüncü 2007). As a quantitative financial model which measures the monetary effect of GCG practice, the proposed financial model uses discounted cash flows (DCF) as the proxy of a company value. Some constraints of GCG practice including regulatory environments and related risks are also carefully identified. A dynamic optimisation model is chosen to capture the fact that the long term corporate financial decision needs to consider the dynamic situation and risks in regard to the company's unique characteristics and business environments. The model will be implemented in a multi-period case so that it represents a

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real world management process which usually takes more than a one year corporate period. The model is purely developed based on financial statements and the equations of the model are developed followed general accepted accounting practice (GAAP), therefore the model shows accounting equations which reflect interaction between income statement, balance sheet and cash flows statements.

As an optimisation model, this project will be applied as a forecasting model. Consequently, the project will be a case study with samples from developing and developed countries. As a forecasting model, the project will be a financial prediction based on the historical data, from the last five years data (i.e. 2005-2009), while the forecasting will be made for the next five years (i.e. from 2010 to 2015). The five-year forecasting will be useful to provide a strategic view for management. For the valuation of the company, the value obtained from the output of the model will be compared with the actual market price. Robustness tests and sensitivity analysis will be conducted to ensure the quality of the model.

The data needed for the project are financial statements and stock prices which can be obtained from the Stock Exchange website. The other data such as the interest rate, discount rate, tax rate and other relevant regulatory environments will be taken from international or national (financial/investment/official) institutions such as the IMF, World Bank, Morgan Stanley, Standard & Poors, taxation department and the Central Bank. For the purpose of forecasting, the project will use Microsoft Excel and an optimisation tool such as Solver.

5. Expected Results and Implications

The proposed model will be an integrated financial optimisation model for achieving good corporate governance. By considering corporate governance practice and regulatory environments such as accounting practice, and financial and accounting market characteristics, the model will assist management to develop sound financial strategies. Finally, management can pursue sound financial strategies to manage the value of a company and achieve the benefits of GCG.

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7. Appendix

Parameters:

CFO_t = net cash flows from operating activities in period *t*;

Int_t = net interest payment (net of tax) in period *t*;

CFI_t = net cash flows from investing activities in period *t*;

CII_t = cash inflows from investing activities in period *t*;

COI_t = cash outflows used by investing activities in period *t*;

$Sales_t$ = net sales in period *t*;

AR_t = account receivables in period t;

AR_{t-1} = account receivables in period $(t-1)$;

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$OREV_t$ = other revenues, i.e. dividend revenue and interest revenue in period *t*;

Rec_t = other receivables in period t;

Rec_{t-1} = other receivables in period $(t-1)$;

$COGS_t$ = cost of good sold in period *t*;

Inv_t = inventories in period t;

Inv_{t-1} = inventories in period $(t-1)$;

AP_t = accounts payable in period t;

AP_{t-1} = accounts payable in period $(t-1)$;

$\overline{OExp_t}$ = adjusted operating expenses from noncash charges in period *t*;

$Prep_t$ = prepaid expenses in period t;

$Prep_{t-1}$ = prepaid expenses in period $(t-1)$;

$Accr_t$ = accrued expenses in period t;

$Accr_{t-1}$ = accrued payable in period $(t-1)$;

Tax_t = current income tax expense in period *t*;

TP_t = tax payable in period t;

TP_{t-1} = tax payable in period $(t-1)$;

FCF_t = free cash flows in period *t*;

$FCFF_t$ = free cash flows to the firm in period *t*;

$FCFE_t$ = free cash flows to equity in period *t*;

Int_t = interest payment (net of tax) in period *t*;

ΔLTD_t = net cash flows from changing in long term debts (excluding DTL) in period *t*;

$\Delta cashPS_t$ =net cash flows from issuing preferred stocks in period *t*;

$pdiv_t$ = cash dividend for preferred stocks in period *t*;

CID_t = cash inflows available from issuing long term debts in period t;

COD_t = cash outflows for retiring long term debts in period t;

CFO_t = net cash flows from operating activities in period *t*;

CFI_t = net cash flows from investing activities in period *t*;

CFF_t = net cash flows from financing activities in period *t*;

div_t = cash dividend for common stockholders in period *t*;

$\Delta cashTS_t$ = net cash from changes in treasury stocks in period *t;*

$\Delta cashCS_t$ = net cash from changes in common stocks in period *t*.

Tax_t = income tax expense in period *t*;

TP_t = income tax payable (current tax expense) in period *t*;

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DTE_t = deferred tax expense in period *t*;

EBT_t = pretax financial income (earnings before tax) in period *t*;

ΔT_t = temporary difference in period *t*;

ΔP_t = permanent difference in period *t*;

exp_t = total expenses in period *t*;

τ = corporate income tax rate;

AR $turnover_t$ = account receivable turnover in period *t*;

$\overline{\text{AR turnover}_t}$ = account receivable turnover of industry in period *t*;

$Sales_t$ = net sales in period *t*;

$\overline{AR_t}$ = net account receivables in period *t*;

ITO_t = inventory turnover in period *t*;

$\overline{ITO_t}$ = inventory turnover of industry in period *t*;

$COGS_t$ = Cost of good sold in period *t*;

Inv_t = inventory (average) in period *t*;

CA_t = current assets in period *t*;

CL_t = current liabilities in period *t*;

Avg $\sum_{t=1}^{T=5} \frac{CA}{CL}$ = average of working capital ratio during previous five year;

$\overline{CA_t}$ = industry current assets in period *t*;

$\overline{CL_t}$ = industry current liabilities in period *t*;

LTD_t = long term debt in period *t*;

$\overline{LTD_t}$ = average long term debt of industry in period *t*;

TD_t = total debt in period *t*;

TA_t = total assets in period *t*;

TD_t = average total debt of industry in period *t*;

$\overline{TA_t}$ = average total assets of industry in period *t*;

DsC_t = debt-service coverage in period *t*;

$\overline{DsC_t}$ = debt-service coverage of the industry in period *t*;

$EBIT_t$ = earnings before interest and taxes in period *t*;

i_t = interest expense in period *t*;

COD_t = cash outflow for debt principals (including financial lease payment and preferred stock dividends) in period *t*;

τ = corporate tax rate;

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α $= 1 +$ dividend growth;

div_t $=$ dividend payment in period *t*;

Q_t= quantity of common stock outstanding in period t;

α_t= the minimum dividend growth in period t;

MV_t $=$ aggregate market value of equity in period *t*;

CIE_t $=$ cash inflows from issuing equity in period *t*;

P_t $=$ market value of common stock per share at the end of period *t*;

EPS_t $=$ earnings per share at the end of period *t*;

$\frac{CIE_t}{Q_t}$ $=$ cash inflow from issuing stock at the end of period *t*;

$\frac{CFI_t}{Q_t}$ $=$ total cash flows for investment activities at the end of period *t*;

k_d $=$ discouting rate;

φSales_t $=$ sales growth in period t;

φPPE_t $=$ PPE growth in period t;

$\overline{\varphi\text{Sales}}$ $=$ average industry's sales growth during period $(t_1 - t_5)$;

$\overline{\varphi PPE}$ $=$ average industry's PPE growth during period $(t_1 - t_5)$;

ΔPPE_t $=$ changes in PPE in period *t*;

$PPE_{(t-1)}$ $=$ amount of PPE in period $(t - 1)$;

ΔSales_t $=$ changes in net sales in period *t*;

$\text{Sales}_{(t-1)}$ $=$ net sales in period $(t - 1)$;

Inc_t $=$ total cash incentives for management in period *t*;

ρ $=$ percentage basis for the management incentives;

$(FCFF_t - FCFF_{(t-1)})$ $=$ an FCFF increase in period *t*.

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